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$$\int \frac{2^x}{1+4^x} dx = \int \frac{2^x dx}{1+2^{2x}}$$

$$\text{Stel } u = 2^x \Rightarrow du = (2^x \ln 2) dx \Rightarrow \frac{1}{\ln 2} du = 2^x dx$$

$$= \frac{1}{\ln 2} \int \frac{du}{1+u^2}$$

$$= \frac{1}{\ln 2} \operatorname{Bgtan}(u) + C$$

$$= \frac{1}{\ln 2} \operatorname{Bgtan}(2^x) + C$$

## References